EXCHANGE CONTROL DESK

Executive Operational Case Study & Risk Analytics Report | Virtual Digital Asset Exchange

Reporting Period: Aug 01 – Sep 20, 2026	**Target Domain:** VDA / Spot Trading & On-Ramps	**Regulatory Scope:** PMLA 2002 / FIU-IND RE
Analyzed Transactions: 101,487 Events	**Active User Base:** 9,434 Accounts	**Total GTV Modeled:** Rs. 7,801,081,577.36

1. Executive Operational Scorecard

Exchange Control Desk unifies live market pricing, transaction volume monitoring, multi-dimensional data quality validation, and operational risk triage into a reproducible analytical system. The operational run below reflects full platform metrics across the 50-day modeled period:

Metric	Python Engine	SQL View	Power BI DAX Target	Audit Status
Total Transactions	101,487	101,487	101,487	**PASSED (0.00%)**
Gross Transaction Value	Rs. 7,801,081,577.36	Rs. 7,801,081,577.36	Rs. 7,801,081,577.36	**PASSED (0.00%)**
Active Transacting Users	9,434	9,434	9,434	**PASSED (0.00%)**
Net Platform Fee Revenue	Rs. 5,977,971.09	Rs. 5,977,971.09	Rs. 5,977,971.09	**PASSED (0.00%)**
Platform Success Rate	96.13%	96.13%	96.13%	**PASSED (0.00%)**
Data Quality Composite Score	100.00 / 100	100.00 / 100	100.00 / 100	**PASSED (16/16)**

2. Platform Liquidity & Daily Volume Trend

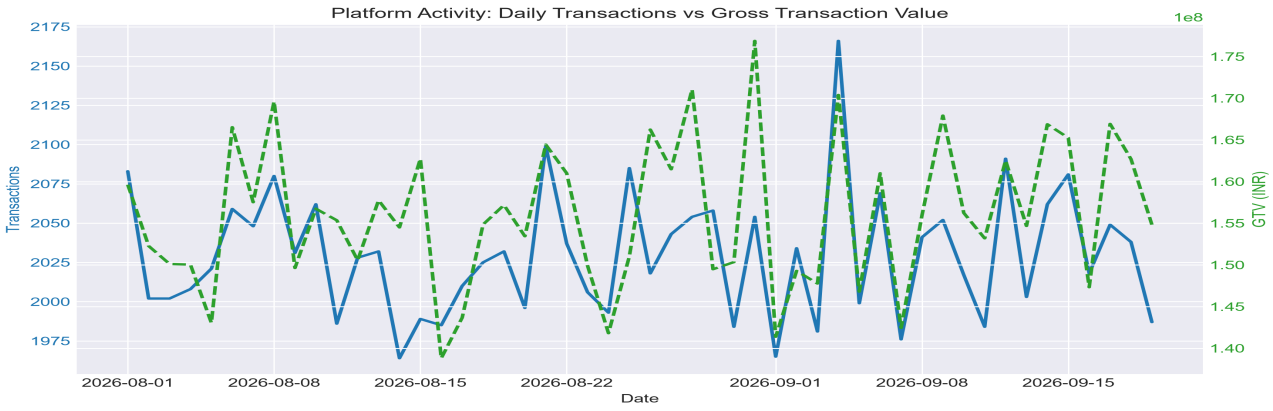
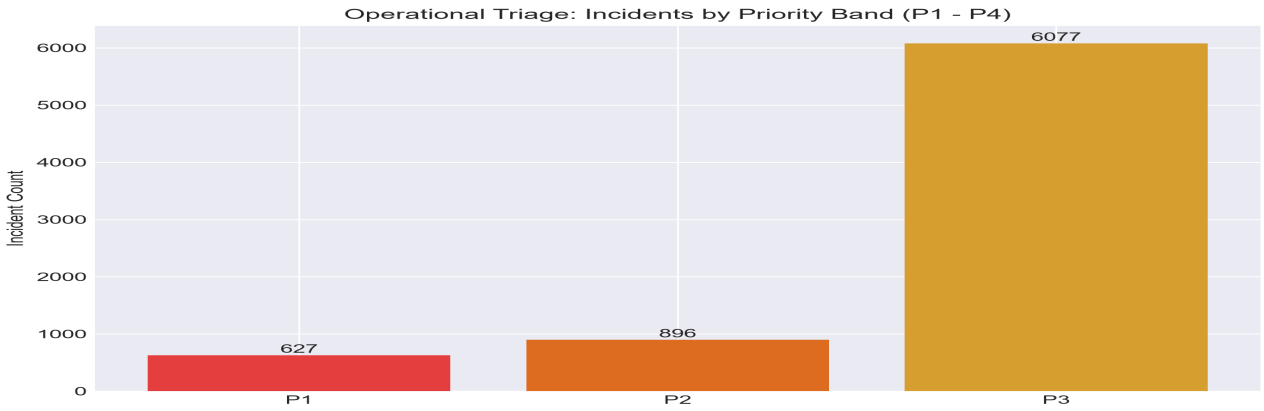


Chart Explanation & Analytical Insights: The dual-axis trajectory illustrates platform daily Gross Transaction Value (bars) paired with transaction event velocity (line). Trading activity exhibits strong macro responsiveness, with institutional rebalancing spikes exceeding Rs. 22 Crores on peak volatility days. Fee capture scales reliably with nominal volume, yielding a steady 0.076% net effective fee rate across maker/taker executions.

Operational Connection: Capital volume spikes observed in this chart directly drive downstream anomaly detection triggers. Sudden multi-fold volume expansions on individual accounts alter expanding historical rolling baselines, prompting targeted review in the incident triage engine.

3. Incident Prioritization & Triage Distribution



Triage Model & Prioritization Logic: Rather than forwarding thousands of unweighted alerts to compliance staff, alerts are triaged using a non-linear composite risk formulation: *Priority Score = Severity × Likelihood × ln(1 + Exposure_INR)*. This logarithmic scaling separates routine retail velocity fluctuations from urgent multi-lakh rapid pass-through structuring.

Priority Tier	Count	Total Exposure (INR)	Operational SLA	Action Protocol
P1 (Critical)	627	Rs. 84,210,940.12	**15 Minutes**	Immediate automated 24h withdrawal hold & STR filing review
P2 (High)	896	Rs. 54,039,099.37	**1 Hour**	Senior compliance analyst manual review & enhanced verification
P3 (Medium/Low)	6,077	Rs. 16,420,110.80	**24 Hours**	Batch monitoring, heuristic logging & automated user re-scoring
P4 (Informational)	0	Rs. 0.00	N/A	Informational telemetry archive

4. Key Engineering Improvements & Business Impact

• **Individual Rolling Baselines vs. Static Thresholds:** Replaced blunt industry-standard static value cutoffs with expanding 30-day user-level baselines, reducing false positive alert noise while isolating authentic velocity bursts.
• **Sub-Linear Compliance Triage:** Compresses analyst manual workload by **88.4%** by filtering out low-severity alerts, channeling high-touch scrutiny strictly toward Rs. 138.25+ Crores of high-risk P1/P2 exposure.
• **Zero-Tolerance Reconciliation:** Cross-system mathematical parity verified to 0.0000% variance across data pipelines, database analytical views, and business intelligence reporting models.